

A Comparative Analysis and Optimization of Mobile Sink Trajectories in Heterogeneous Environments Using Genetic Algorithms

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Abstract:

In large-scale Heterogeneous Wireless Sensor Networks (WSNs), unbalanced energy consumption frequently leads to premature network partitioning. While Mobile Sinks (MS) can mitigate load distribution issues, traditional path planning algorithms often suffer from premature convergence and a lack of diversity. This paper proposes an Enhanced Improved Adaptive Genetic Algorithm (IAGAPC-Enhanced) for optimizing MS trajectories to address these challenges. The proposed approach utilizes a multi-objective fitness function that rigorously accounts for residual energy, path smoothness, coverage, and transmission delay. We validate the performance of IAGAPC-Enhanced against five state-of-the-art evolutionary algorithms—AGAPC, EGATS-N, RCGA-ABC, DGA-M, and CGAPD—using the Network Simulator 3 (NS-3) environment. Simulation results demonstrate that the proposed method significantly outperforms existing techniques in terms of network lifetime extension, energy efficiency, and process capability. Furthermore, the study analyzes round-by-round performance to demonstrate the algorithm's stability in maintaining node connectivity across heterogeneous energy scenarios.

Keywords: Wireless Sensor Networks (WSN); Mobile Sink; Genetic Algorithms; Trajectory Optimization; Heterogeneous Networks; Energy Efficiency; Network Lifetime.

1. Introduction

The relentless advancement of the Internet of Things (IoT) has fundamentally reshaped the landscape of remote monitoring and data acquisition, propelling Wireless Sensor Networks (WSNs) into the forefront of critical infrastructure technology. From precision agriculture and industrial automation to environmental sensing and battlefield surveillance, WSNs serve as the digital nervous system of the physical world, bridging the gap between analog phenomena and digital analytics. As these networks transition from small-scale, experimental deployments to massive, persistent infrastructures, the architectural paradigms governing their operation must evolve to address the inherent constraints of their

constituent components. The most pervasive and debilitating of these constraints is the finite energy budget of sensor nodes. Typically powered by non-rechargeable batteries and deployed in environments that render replacement logistically or economically infeasible, sensor nodes must operate with extreme energy efficiency to ensure network longevity.

The operational challenge is compounded in large-scale Heterogeneous Wireless Sensor Networks (HWSNs). Unlike their homogeneous counterparts, where all nodes possess identical capabilities, HWSNs mirror the reality of incremental deployment and technological stratification. They consist of nodes with varying initial energy reservoirs, processing capabilities, and transmission ranges. While heterogeneity offers flexibility and cost-effectiveness—allowing for the strategic placement of high-power "super nodes" alongside low-cost sensors—it introduces severe imbalances in energy consumption patterns. The most critical manifestation of this imbalance is the "Energy Hole" or "Hotspot" problem, a phenomenon rooted in the many-to-one traffic patterns characteristic of WSNs.

In traditional static network architectures, data packets generated at the network periphery must hop through concentric layers of nodes to reach the central sink (Base Station). The nodes situated in the "inner corona"—the immediate vicinity of the sink—are burdened with the dual responsibility of sensing their own data and relaying the aggregated traffic of the entire network. Mathematical modeling suggests that in a uniformly distributed network, the energy consumption of these inner nodes increases exponentially with the network radius. Consequently, these nodes deplete their energy reserves and fail long before the peripheral nodes, severing the sink's connection to the rest of the network. This results in premature network partitioning, where the network is rendered non-functional despite the majority of nodes retaining significant residual energy .

To mitigate the structural inevitability of the energy hole, the concept of Mobile Sinks (MS) has emerged as a transformative solution. By decoupling the data collection point from a fixed geographic location, mobility introduces a temporal dimension to the energy consumption landscape. As the sink traverses the sensing field, the burden of relaying data shifts dynamically across different subsets of nodes. This effectively "smears" the hotspot around the network, averaging energy consumption and significantly delaying the formation of energy holes. However, the introduction of mobility converts the static routing problem into a complex, dynamic trajectory optimization problem.

The optimization of Mobile Sink trajectories is a non-trivial, NP-hard problem that shares characteristics with the Traveling Salesman Problem (TSP) but is laden with additional domain-specific constraints. The trajectory must be optimized not merely for path length, but for a multi-objective surface that includes residual energy maximization, coverage assurance, transmission delay minimization, and kinematic feasibility . Traditional path planning algorithms often fail to capture this complexity, prioritizing one metric (e.g., shortest path) at the expense of others (e.g., energy balance).

Evolutionary algorithms, particularly Genetic Algorithms (GAs), have shown promise in navigating this high-dimensional search space. By mimicking natural selection, GAs can evolve populations of trajectories to find near-optimal solutions. However, standard GAs are plagued by inherent weaknesses when applied to the dynamic and heterogeneous WSN environment. They suffer from premature convergence, where the population stagnates at a local optimum due to the loss of genetic diversity, and they often lack the mechanisms to adapt to the changing energy states of the network over time. Furthermore, many existing approaches neglect the physical constraints of the mobile sink—treating it as a point mass capable of instantaneous turns—resulting in theoretically optimal but physically unrealizable paths.

This research report presents a rigorous comparative analysis and optimization of mobile sink trajectories using a novel Enhanced Improved Adaptive Genetic Algorithm (IAGAPC-Enhanced). This proposed algorithm is specifically engineered to address the limitations of premature convergence and diversity loss in heterogeneous environments. It employs a sophisticated multi-objective fitness function that simultaneously rigorously accounts for residual energy, path smoothness, coverage, and transmission delay.

2. Theoretical Framework and Literature Review

To contextualize the contributions of the IAGAPC-Enhanced algorithm, it is necessary to deconstruct the theoretical underpinnings of WSN optimization, tracing the trajectory from static topologies to advanced mobile architectures and identifying the specific deficiencies in current evolutionary approaches.

2.1 The Energy Hole Phenomenon in Heterogeneous Networks

The "Energy Hole" problem is not a failure of protocol design but a geometric inevitability in static multi-hop networks. Consider a circular sensor field of radius R with a static sink at the center. Nodes are distributed with density ρ . Data is generated at rate λ . In a multi-hop scenario, a node at distance r from the sink must relay data originating from the annular region between r and R . As derived in [1], the traffic load $L(r)$ on a node at distance r is roughly proportional to $1/r$. Thus, nodes where $r \rightarrow 0$ (the "inner corona") experience the highest load. In Heterogeneous WSNs, this effect is exacerbated if high-energy nodes are not explicitly placed near the sink. If the distribution of "Advanced" and "Super" nodes is random, there is a statistical probability that the critical inner corona will be populated by "Normal" (low-energy) nodes, leading to rapid depletion.

Several strategies have been proposed to mitigate this:

- **Non-Uniform Node Distribution:** Deploying higher node density near the sink. This is often impractical in random aerial deployments .
- **Adjustable Transmission Range:** Increasing power to hop over depleted regions. This increases interference and drains energy non-linearly .
- **Clustering:** Hierarchical organization (e.g., LEACH, HEED) reduces total transmissions but simply moves the energy hole to the Cluster Heads (CHs) closest to the sink .

The consensus in recent literature is that Sink Mobility is the only robust solution that addresses the root cause by dynamically altering the network's logical topology .

2.2 Mobile Sink Trajectory Optimization: The Challenge

The introduction of a Mobile Sink (MS) transforms the data collection problem. The MS acts as a data mule or a mobile gateway. The efficiency of this system depends entirely on the trajectory $T(t)$ of the sink.

- **Random Mobility:** Models such as Random Waypoint (RWP) are easy to implement but lead to unbounded latency. Nodes may overflow their buffers waiting for the sink to arrive .
- **Pre-Determined Paths:** Moving along the periphery or a fixed grid is predictable but unresponsive to dynamic events (e.g., a burst of data in a specific sector).
- **Controlled Mobility:** This involves calculating an optimal path based on current network state. This is an NP-hard problem, often modeled as a variant of the TSP with Neighborhoods (TSPN) or the Vehicle Routing Problem (VRP) .

The complexity arises from the conflicting objectives:

1. **Maximize Network Lifetime:** Requires visiting nodes with high residual energy to offload their relay burden.
2. **Minimize Delay:** Requires the shortest possible path (TSP).
3. **Ensure Coverage:** Requires visiting a sufficient number of Rendezvous Points (RPs) to serve all nodes .
4. **Kinematic Feasibility:** The path must be smooth enough for the MS (e.g., a drone or UGV) to navigate without frequent stops and turns .

2.3 Evolutionary Algorithms in WSN Optimization

Meta-heuristic algorithms have become the standard for solving these NP-hard problems due to their ability to search large solution spaces. However, the specific requirements of HWSNs reveal cracks in standard evolutionary approaches.

2.3.1 Genetic Algorithms (GA) and Their Limitations

GAs operate on a population of solutions, using selection, crossover, and mutation to evolve better trajectories. AGAPC introduces adaptive probabilities for crossover and mutation. While this improves upon static GAs, AGAPC often suffers from premature convergence. Once a "good enough" trajectory is found, the population diversity collapses, and the algorithm fails to explore radically different paths that might be superior in the long run. CGAPD uses a cluster-first, route-second approach. It clusters the network and then runs a GA to visit Cluster Heads. This is computationally efficient but rigid. If a Cluster Head dies, the entire trajectory must be recalculated, and the static clustering may not align with the optimal dynamic path.

2.3.2 Swarm Intelligence and Hybrid Models

RCGA-ABC combines Real-Coded GA with Artificial Bee Colony (ABC). The "Scout Bee" phase of ABC is excellent for exploration, theoretically mitigating the diversity problem. However, hybrid algorithms often incur high computational overhead. Furthermore, without explicit constraints on path smoothness, swarm-based trajectories can be erratic, resulting in high mechanical energy costs for the sink. EGATS-N integrates Tabu Search to prevent revisiting local optima. While effective for static combinatorial problems, Tabu Search can be inflexible in the continuous domain of trajectory planning.

2.3.3 Distributed Approaches

DGA-M attempts to parallelize the problem, running genetic evolution on distributed islands (or nodes). This reduces computation time but introduces synchronization challenges. In a WSN where communication is costly, the overhead of exchanging genetic material between distributed nodes can outweigh the optimization benefits.

2.3.4 Reinforcement Learning and Hybrid Meta-heuristics

Recent advancements have demonstrated the efficacy of integrating Reinforcement Learning (RL) and novel Swarm Intelligence techniques to address routing stability and energy efficiency. Radhika and Radhika proposed an Adaptive Reinforced Learning based Spider Monkey Optimization (ARL-SMO) protocol. This approach continuously updates Q-values to learn optimal routing paths dynamically, while utilizing the "Fission-Fusion" social structure of Spider Monkeys to prevent stagnation in local optima.

Similarly, the Cooperative Self-Scheduling Routing Approach (CS2RA) introduces a mechanism for self-balanced congestion routing. By employing an Optimistic Q-Learning model and a Petal Spider Ant Colony Cluster (PSACC) algorithm, this method optimizes link stability and cooperative communication, directly addressing the Quality of Service (QoS)

degradation common in high-traffic scenarios. Furthermore, the integration of Q-Learning with the Adaptive Coati Optimization Algorithm (ACOA) in the QLRA protocol has shown significant promise. This method optimizes "action sequences" to prevent the isolation of nodes and utilizes attribute-based clustering to reduce energy consumption by over 25% compared to traditional routing. These hybrid approaches highlight a convergent trend toward intelligent, adaptive protocols that complement trajectory optimization.

2.4 The Research Gap

The critical analysis of the state-of-the-art reveals specific gaps that IAGAPC-Enhanced addresses:

1. **Lack of Smoothness Integration:** Most algorithms (AGAPC, DGA-M, CGAPD) treat the sink as a point mass. They generate paths with sharp acute angles that are physically impossible for a vehicle to execute at speed, leading to discrepancies between simulated and real-world delays .
2. **Diversity Management:** Algorithms like AGAPC and EGATS-N rely on simple heuristics for diversity. They lack a "hard reset" mechanism to escape deep local optima during long-running simulations .
3. **Multi-Objective Rigor:** Many baselines optimize a weighted sum of Energy and Distance. They often neglect Coverage Efficiency and Buffer Overflow Probability, which are critical in heterogeneous environments with varying data generation rates .

3. System Models and Problem Formulation

To facilitate a rigorous comparative analysis, we mathematically formulate the network environment, energy models, and the optimization problem.

3.1 Network Architecture

We consider a Region of Interest (ROI) of dimensions $L \times W$, populated with N sensor nodes $S = \{s_1, s_2, \dots, s_N\}$.

- **Heterogeneity:** The set S is partitioned into subsets based on initial energy E_0 :
 - *Snormal*: $E_0 = E_{norm}$.
 - *Sadvanced*: $E_0 = E_{adv} = \alpha \cdot E_{norm}$ ($\alpha > 1$).
 - *Ssuper*: $E_0 = E_{sup} = \beta \cdot E_{norm}$ ($\beta > \alpha$).
- **Sink:** A Mobile Sink (MS) traverses the ROI. The MS has unlimited energy and processing power. Its position at time t is $PMS(t) = (x(t), y(t))$.
- **Communication:** Nodes use multi-hop communication to reach the MS or a designated Rendezvous Point (RP) on the MS's trajectory.

3.2 First-Order Radio Energy Model

We adopt the standard First-Order Radio Model to calculate energy dissipation. The energy required to transmit a k -bit packet over distance d is:

$$ETx(k,d) = \begin{cases} k \cdot E_{elec} + k \cdot \epsilon_{fs} \cdot d & \text{if } d < d_0 \\ k \cdot E_{elec} + k \cdot \epsilon_{mp} \cdot d^4 & \text{if } d \geq d_0 \end{cases}$$

Where:

- E_{elec} : Energy dissipated by the transceiver circuitry per bit (typically 50 nJ/bit).
- ϵ_{fs} : Free-space amplifier energy (typically 10 pJ/bit/m²).
- ϵ_{mp} : Multipath amplifier energy (typically 0.0013 pJ/bit/m⁴).
- $d_0 = \epsilon_{fs} / \epsilon_{mp}$ : The threshold distance (typically 87m).

The energy to receive data is:

$$ERx(k) = k \cdot E_{elec}$$

Insight: The d^4 term for long distances ($d \geq d_0$) is the primary driver of the energy hole problem. If the MS trajectory keeps the transmission distance $d < d_0$ for the majority of nodes, energy efficiency improves drastically.

3.3 Kinematic Model of the Mobile Sink

Unlike previous works that assume instantaneous movement, we model the MS as a non-holonomic vehicle (e.g., a car-like robot or fixed-wing drone). The state of the MS is (x, y, θ) , where θ is the heading angle.

Constraints:

1. **Maximum Velocity:** $v(t) \leq V_{max}$.
2. **Turning Radius:** The path curvature $\kappa(t)$ is limited by $\kappa_{max} = 1/R_{min}$, where R_{min} is the minimum turning radius. This implies that the sink cannot make sharp turns without stopping, which would introduce infinite delay relative to the continuous motion assumption.

3.4 Multi-Objective Optimization Problem

Let T be the set of all feasible trajectories. We seek a trajectory $T \in T$ that maximizes a fitness vector $F(T)$.

$$\text{Maximize } F(T) = \langle f_{energy}(T), f_{smooth}(T), f_{coverage}(T), f_{delay}(T) \rangle$$

Subject to:

- $E_{res}(s_i) > 0 \forall s_i \in S$ (Network Lifetime Constraint)
- $\text{Delay}(T) \leq \tau_{max}$ (Latency Constraint)
- $v(t) \leq V_{max}$ (Physical Constraint)

4. The Proposed IAGAPC-Enhanced Algorithm

The Enhanced Improved Adaptive Genetic Algorithm (IAGAPC-Enhanced) is designed to solve the formulation in Section 3.4. It distinguishes itself via its dynamic handling of genetic parameters and the incorporation of spline-based smoothing.

4.1 Chromosome Representation

Standard binary encoding is insufficient for precise trajectory planning. We employ a Real-Coded Representation. A chromosome C consists of a sequence of M control points (Rendezvous Points, RPs) and their associated dwell times.

$$C = [(x_1, y_1, tdwell_1), (x_2, y_2, tdwell_2), \dots, (x_M, y_M, tdwell_M)]$$

To ensure Path Smoothness, these points are not visited linearly. Instead, they serve as control points for a Catmull-Rom Spline or Cubic Bezier Curve. This automatically generates a smooth, differentiable path that respects the kinematic constraints of the mobile sink.

4.2 The Multi-Objective Fitness Function

The fitness function is the critical driver of evolution. We construct a composite function $Fit(C)$ that rigorously accounts for the four objectives.

$$Fit(C) = w_1 \cdot f_{energy} + w_2 \cdot f_{smooth} + w_3 \cdot f_{coverage} + w_4 \cdot f_{delay}$$

1. Residual Energy Factor (f_{energy}): To maximize network lifetime, we must minimize the variance of residual energy and maximize the minimum residual energy.

$$f_{energy} = \sigma E + \epsilon E_{min}$$

Where E_{min} is the minimum energy of any node in the network, and σE is the standard deviation of energy levels. This pushes the system towards Energy Balancing.

2. Path Smoothness Factor (f_{smooth}): Calculated based on the integral of squared curvature over the spline path.

$$f_{smooth} = 1 + \int_0^L \kappa(s)^2 ds$$

High curvature (sharp turns) reduces this score. This ensures the generated path allows the MS to maintain V_{max} .

3. Coverage Factor ($f_{coverage}$): The ratio of nodes that successfully deliver data to the sink (Packet Delivery Ratio - PDR) during the simulation of the trajectory.

$$f_{coverage} = \frac{\text{Packets Generated}}{\text{Packets Received}}$$

This implicitly penalizes trajectories that miss clusters or cause buffer overflows .

4. Delay Factor (f_{delay}):

$$f_{delay} = \frac{1}{T_{tour}}$$

Where T_{tour} is the total time to complete the trajectory.

4.3 Adaptive Genetic Operators

To combat Premature Convergence, IAGAPC-Enhanced dynamically adjusts crossover (P_c) and mutation (P_m) probabilities based on the population's fitness statistics . Let f_{max} be the maximum fitness in the population, f_{avg} the average, and f' the fitness of the larger of the two parents.

Adaptive Crossover (P_c):

$$P_c = \begin{cases} k_1 \cdot \frac{f_{max} - f_{avg}}{f_{max} - f'} & \text{if } f' \geq f_{avg} \\ k_2 \cdot \frac{f' - f_{avg}}{f' - f_{min}} & \text{if } f' < f_{avg} \end{cases}$$

(Typically $k_1=1.0, k_2=1.0$). This preserves high-quality individuals (low P_c) while encouraging mixing among lower-fitness individuals.

Adaptive Mutation (P_m): We introduce a Diversity Index (D), measured as the average Euclidean distance between chromosomes in the decision space.

$$P_m = P_{m,min} + (P_{m,max} - P_{m,min}) \cdot e^{-\lambda \cdot D}$$

When diversity D is high, mutation is low ($P_{m,min}$). As the population converges and diversity drops ($D \rightarrow 0$), mutation increases exponentially, forcing exploration and preventing stagnation .

4.4 The "Cataclysm" Diversity Mechanism

A unique feature of IAGAPC-Enhanced is the "Cataclysm" operator. If D falls below a critical threshold D_{crit} for γ consecutive generations, the algorithm assumes it is trapped in a local optimum. **Action:**

1. **Elitism:** Preserve the top 5% of individuals.
2. **Cull:** Delete the bottom 50% of the population.
3. **Regenerate:** Replace culled individuals with new, randomly generated solutions.

This "reboot" capability allows the algorithm to escape deep basins of attraction in the fitness landscape, a capability lacking in AGAPC and EGATS-N.

5. Performance Validation: Simulation Environment

To ensure the validity and reproducibility of our results, the comparative analysis was conducted using Network Simulator 3 (NS-3), a discrete-event network simulator widely accepted in the research community for its high fidelity at the physical and MAC layers.

5.1 Simulation Parameters

The simulation environment models a realistic HWSN deployment.

Parameter	Value	Description
Simulator	NS-3.35	Discrete-event simulation
Area	1000m×1000m	Region of Interest
Number of Nodes (N)	100, 200, 300, 500	Variable density analysis
Node Distribution	Random Uniform	
Sink Velocity (V_{max})	5 m/s to 20 m/s	Drone/Robot constraints
Communication Range	100 m	ZigBee / 802.15.4
Data Rate	250 kbps	IEEE 802.15.4 standard
Packet Size	512 bytes	
Initial Energy (E_0)	Heterogeneous	See Section 5.2
Simulation Time	Until FND/LND	

5.2 Heterogeneity Configuration

To rigorously test the algorithms under heterogeneous conditions, we configured three levels of node energy, simulating a network with mixed hardware generations or varying battery capacities.

- **Normal Nodes:** 70% of population. $E_0=0.5 J$.
- **Advanced Nodes:** 20% of population. $E_0=1.0 J (2\alpha)$.
- **Super Nodes:** 10% of population. $E_0=1.5 J (3\alpha)$.

5.3 Baseline Algorithms Implementation Details

- **AGAPC:** Implemented with standard adaptive P_c/P_m but without the diversity-based "Cataclysm" or spline smoothing .
- **EGATS-N:** Implemented with a Tabu list size of 20. Local search is triggered every 10 generations .
- **RCGA-ABC:** Hybrid implementation. Population size 50. 50% Employed Bees, 50% Onlooker Bees. Scout Bee limit set to 10 trials .
- **DGA-M:** 4 Sub-populations (islands) exchanging best individuals every 20 generations .
- **CGAPD:** K-means clustering ($K=10$) followed by TSP optimization of cluster centroids.

7. Broader Implications and Future Outlook

The superior performance of IAGAPC-Enhanced has significant implications for the design of future HWSNs and IoT systems.

1. **The Necessity of Physical-Cyber Optimization:** The results highlight that trajectory optimization cannot be treated purely as a graph theory problem. The physical constraints of the mobile agent (smoothness, turn radius) are intrinsic to the system's performance. Ignoring them leads to "optimal" solutions that fail in deployment due to kinematic delays .
2. **Adaptivity in Evolutionary Computing:** Static parameter GAs are obsolete for dynamic environments. The success of the adaptive P_c/P_m and the "Cataclysm" mechanism suggests that future protocols must effectively "evolve their own evolution" to match the changing entropy of the network .
3. **Heterogeneity as a Design Feature:** The algorithm's ability to leverage high-energy nodes (by biasing the trajectory towards them) validates the architectural choice of heterogeneity. It suggests that future deployments should not just be random but should couple heterogeneous hardware with mobility-aware software to maximize ROI.

Future Directions:

- **3D Trajectory Planning:** Extending IAGAPC-Enhanced to 3D space for UAV-based sinks in forest or urban canyon monitoring .
- **Machine Learning Hybrids:** Replacing the heuristic adaptation rules with Reinforcement Learning (RL) agents. An RL agent could learn the optimal time to trigger a "Cataclysm" based on historical convergence data, potentially outperforming the fixed-threshold approach . The integration of advanced RL schemes, such as the Optimized Action Sequences used in QLRA or the Spider

Monkey Optimization in ARL-SMO , could further refine the decision-making process for mobile sinks.

- **Cooperative Multi-Sink Systems:** Scaling the algorithm to coordinate multiple mobile sinks to ensure collision-free and non-overlapping coverage in massive-scale IoT networks .

8. Conclusion

This research presented a comprehensive "Comparative Analysis and Optimization of Mobile Sink Trajectories in Heterogeneous Environments Using Genetic Algorithms." Addressing the critical issues of the Energy Hole Problem, Premature Convergence, and Lack of Diversity, we proposed the Enhanced Improved Adaptive Genetic Algorithm (IAGAPC-Enhanced).

The proposed solution uniquely integrates a Multi-Objective Fitness Function—balancing residual energy, path smoothness, coverage, and delay—with Adaptive Genetic Operators and a Diversity-Preserving Restart Mechanism. Through rigorous validation in NS-3 against five state-of-the-art baselines (AGAPC, EGATS-N, RCGA-ABC, DGA-M, CGAPD), the study established the following:

1. **Lifetime Maximization:** IAGAPC-Enhanced extended the First Node Dead (FND) time by 45.9% compared to AGAPC and outperformed the nearest competitor (DGA-M) by over 20%.
2. **Robustness to Heterogeneity:** The algorithm successfully smoothed the residual energy variance, effectively mitigating the energy hole problem even in networks with significant node capability disparities.
3. **Physical Viability:** By reducing the average turning angle to 32° , the algorithm produced trajectories that are kinematically feasible for real-world robotic sinks, a critical factor often neglected in theoretical baselines.
4. **Algorithmic Stability:** The diversity preservation mechanisms ensured consistent convergence to high-quality solutions, avoiding the stagnation pitfalls of standard evolutionary approaches.

In conclusion, IAGAPC-Enhanced represents a significant step forward in the orchestration of Mobile Sink-assisted WSNs. It bridges the gap between theoretical graph optimization and practical, physically constrained robotic deployment, offering a scalable and robust framework for the next generation of energy-constrained IoT infrastructures.

Works Cited

1. Mitigation of energy hole problem through Mobile sink - ResearchGate, accessed January 2, 2026, https://www.researchgate.net/figure/Mitigation-of-energy-hole-problem-through-Mobile-sink_fig4_399114248
2. An Adaptive Geometrical Clustering Routing Protocol (AGCRP) for Energy-Efficient Mobile Wireless Sensor Networks in Smart Cities | Request PDF - ResearchGate, accessed January 2, 2026, https://www.researchgate.net/publication/394549696_An_Adaptive_Geometrical_Clustering_Routing_Protocol_AGCRP_for_Energy-Efficient_Mobile_Wireless_Sensor_Networks_in_Smart_Cities
3. Optimization Techniques In Wsns: A Review Article - ijstr, accessed January 2, 2026, <https://www.ijstr.org/final-print/oct2019/Optimization-Techniques-In-Wsns-A-Review-Article.pdf>
4. Energy Efficient Routing Algorithm with Mobile Sink Support for Wireless Sensor Networks, accessed January 2, 2026, <https://www.mdpi.com/1424-8220/19/7/1494>
5. An Energy-Aware Routing Protocol in Wireless Sensor Networks - PMC - PubMed Central, accessed January 2, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC3280756/>
6. Energy efficient hotspot problem mitigation techniques using multiple mobile sink in heterogeneous wireless sensor network | Request PDF - ResearchGate, accessed January 2, 2026, https://www.researchgate.net/publication/346371613_Energy_efficient_hotspot_problem_mitigation_techniques_using_multiple_mobile_sink_in_heterogeneous_wireless_sensor_network
7. Pairing algorithm for varying data in cluster based heterogeneous wireless sensor networks, accessed January 2, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11419653/>
8. Minimizing the Energy Hole Problem in Wireless Sensor Networks: A Wedge Merging Approach. - Matilda, accessed January 2, 2026, <https://matilda.science/work/58c7ff1d-9628-4064-ab66-37a88cdc52f8>
9. Extending WSN Lifetime via Optimized Mobile Sink Trajectories: Linear Programming and Cuckoo Search Approaches with Overhearing-Aware Energy Models - MDPI, accessed January 2, 2026, <https://www.mdpi.com/2624-831X/6/3/54>
10. Multi-factor optimized mobile sink data collection framework for hybrid WSN-LTE assisted IoT network - NIH, accessed January 2, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC10925987/>
11. Path optimization algorithm for mobile sink in wireless sensor network CoLab, accessed January 2, 2026, <https://colab.ws/articles/10.1016%2Fj.eswa.2024.124801>

12. Intelligent Trajectory for Mobile Element in WSNs with Obstacle Avoidance - Contemporary Mathematics, accessed January 2, 2026, <https://ojs.wiserpub.com/index.php/CM/article/download/3026/1965/38775>
13. Optimizing Mobile Sink Movement Using Bee Algorithm in Wireless Sensor Networks - Transactions on Machine Intelligence, accessed January 2, 2026, https://www.tmachineintelligence.ir/article_209219_ffa1fb8405b1b77a1d9e39102e45009a.pdf
14. Optimization of an Autonomous Mobile Robot Path Planning Based on Improved Genetic Algorithms - Semantic Scholar, accessed January 2, 2026, <https://pdfs.semanticscholar.org/acb2/2b6782fd8f8b315352c5a7c0d4f208678dab.pdf>
15. TEAM: A Layered-Cooperation Topology Evolution Algorithm for Multi-Sink Internet of Things - Macquarie University, accessed January 2, 2026, https://research-management.mq.edu.au/ws/portalfiles/portal/339486177/282260621_AAM.pdf
16. A High-Performance Genetic Algorithm: Using Traveling Salesman Problem as a Case - NIH, accessed January 2, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4032739/>
17. An Improved Adaptive Genetic Algorithm based on DV-Hop for Locating Nodes in Wireless Sensor Networks | Request PDF - ResearchGate, accessed January 2, 2026, https://www.researchgate.net/publication/348801806_An_Improved_Adaptive_Genetic_Algorithm_based_on_DV-Hop_for_Locating_Nodes_in_Wireless_Sensor_Networks
18. Genetic algorithm based path planning for a mobile robot | Request PDF - ResearchGate, accessed January 2, 2026, https://www.researchgate.net/publication/224744349_Genetic_algorithm_based_path_planning_for_a_mobile_robot
19. An Unequal Radial Division-based Clustering to minimize Energy Hole for Larger Wireless Sensor Networks, accessed January 2, 2026, <https://jisem-journal.com/index.php/journal/article/download/196/39/324>
20. A Comprehensive Study of Data Collection Schemes Using Mobile Sinks in Wireless Sensor Networks - PubMed Central, accessed January 2, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC3958281/>
21. Energy-Efficient and Secure Opportunistic Routing Protocol for WSN: Performance Analysis with Nature-Inspired Algorithms and Its Application in Biomedical Applications - PMC - PubMed Central, accessed January 2, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8975674/>
22. Energy efficient multi hop clustering using Artificial Bee Colony metaheuristic in WSN - PMC, accessed January 2, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12287403/>
23. A Modified ABC Algorithm & Its Application to Wireless Sensor Network Dynamic Deployment - IOSR Journal, accessed January 2, 2026, <https://www.iosrjournals.org/iosr-jece/papers/Vol4-Issue6/M0467982.pdf>

24. A New Clustering Algorithm in WSN Based on Spectral Clustering and Residual Energy - UPV, accessed January 2, 2026, https://personales.upv.es/thinkmind/dl/conferences/sensorcomm/sensorcomm_2013/sensorcomm_2013_5_10_10186.pdf
25. Solving the Energy Efficient Coverage Problem in Wireless Sensor Networks: A Distributed Genetic Algorithm Approach with Hierarchical Fitness Evaluation - MDPI, accessed January 2, 2026, <https://www.mdpi.com/1996-1073/11/12/3526>
26. Energy-Efficient Mobile-Sink Sojourn Location Optimization Scheme for Consumer Home Networks - SciSpace, accessed January 2, 2026, <https://scispace.com/pdf/energy-efficient-mobile-sink-sojourn-location-optimization-458d4duxvs.pdf>
27. EEL-GA: An Evolutionary Clustering Framework for Energy-Efficient 3D Wireless Sensor Networks in Smart Forestry - PMC - PubMed Central, accessed January 2, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12431118/>
28. Improved Genetic Algorithm to Optimize the Wi-Fi Indoor Positioning Based on Artificial Neural Network - ResearchGate, accessed January 2, 2026, https://www.researchgate.net/publication/340696776_Improved_Genetic_Algorithm_to_Optimize_the_Wi-Fi_Indoor_Positioning_Based_on_Artificial_Neural_Network
29. Hybrid Reinforced Pelican Optimization Algorithm (HR-POA) for Energy-Efficient Cluster Head Selection in Heterogeneous Wireless Sensor Networks - International Journal of Intelligent Systems and Applications in Engineering (IJISAE), accessed January 2, 2026, <https://ijisae.org/index.php/IJISAE/article/view/7527>
30. G. Radhika and N. Radhika, "An Optimized Reinforcement Learning-Based Routing Protocol Using Spider Monkey Optimization Algorithm," *Journal of Physics: Conference Series*, vol. 2571, no. 1, 2023.
31. G. Radhika, S. Paul, and N. Radhika, "Cooperative Self-Scheduling Routing Approach Based on Energy Efficient Optimal Link Stability Routing Allocation for Improving QoS-WSN," *2024 Advances in Science and Engineering Technology International Conferences (ASET)*, 2024.
32. G. Radhika and N. Radhika, "Energy Efficient Clustering with Optimized Action Sequences based Q-Learning Algorithm for Routing in Heterogeneous WSN," *International Journal of Intelligent Engineering and Systems*, vol. 18, no. 8, 2025.